Department of Computer Science and Engineering

Assignment Questions

Course Code & Name: ITA03 - Mobile Computing

Assignment Title:	**Design of a Rapidly Deployable Mobile Communication Network for Post-Disaster Emergency Response**
Course Outcome(s) - CO:	CO1: Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems. CO2: Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications. CO3: Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.
Bloom's Taxonomy Level	L3 - Apply; L4 - Analyse
SDG Mapping	SDG 9 - Industry, Innovation and Infrastructure; SDG 11 - Sustainable Cities and Communities; SDG 13 - Climate Action

Problem Statement

A coastal district has just been struck by a major cyclone that has knocked out the region's fixed telecom towers and fibre backhaul, cutting off several villages and a relief camp from the rest of the network. A disaster-management team must rapidly re-establish mobile connectivity to coordinate rescue operations, track relief vehicles, and let affected residents reach emergency services. The proposed work analyzes cellular components and protocols, compares multiple access techniques and network evolution, and designs both a Mobile IP based mobility scheme and an infrastructure-less ad hoc routing scheme while evaluating reliability, delay and resilience to network partitioning.

Report Overview

This report presents a practical hybrid communication architecture for restoring connectivity after a major disaster. The solution combines a portable cellular base station, resilient backhaul, Mobile IP based mobility support and an AODV-based ad hoc mesh to maintain communication even when conventional infrastructure is damaged.

Key Design Goals

rapid deployment, emergency traffic prioritization, mobility support, low communication delay, reliability and resilience to network partitioning.

1. Introduction

During a major cyclone or other natural disaster, fixed communication towers, power systems and fibre backhaul can fail simultaneously. Rescue teams then lose coordination capability, affected residents cannot contact emergency services, and relief operations become difficult to manage. A rapidly deployable mobile communication network is therefore required to restore essential connectivity within a short time.

The proposed solution uses a hybrid architecture rather than depending on a single technology. A portable Cell-on-Wheels (COW) provides temporary cellular coverage, satellite or microwave links provide backhaul, Mobile IP supports movement across network segments, and an ad hoc mesh extends communication into isolated villages and relief camps.

2. Scenario Analysis and Design Assumptions

The disaster zone consists of damaged villages, a central relief camp, moving rescue teams and a temporary command centre. Some external network infrastructure may survive outside the affected area, but the local cellular towers and fibre links are assumed to be unavailable. Communication must therefore be restored using portable and battery-supported equipment.

Parameter	**Design Assumption**
Coverage area	Approximately 3–5 km around the temporary cellular deployment
Users	Survivors, rescue teams, medical staff and disaster-management personnel
Traffic	Priority voice, SMS alerts, location updates and coordination data
Backhaul	Satellite or microwave; surviving terrestrial connectivity when available
Power	Battery and generator support with optional solar backup
Isolated areas	Relief camps and villages use a multi-hop ad hoc mesh

3. Proposed Network Architecture

The architecture below is the central design of the proposed system. It integrates temporary cellular infrastructure with alternative backhaul and an ad hoc extension. The architecture is designed so that loss of one communication path does not completely isolate all users.

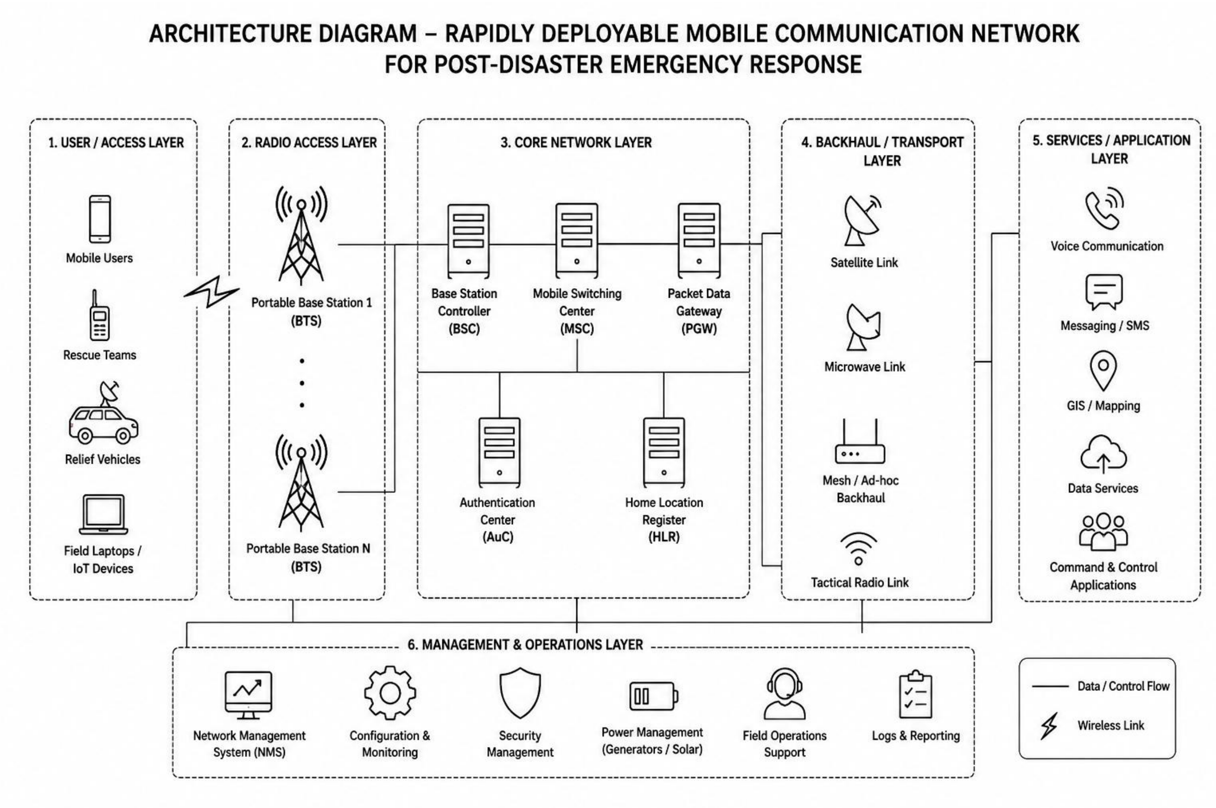


Figure 1. Proposed rapidly deployable post-disaster mobile communication architecture

The proposed architecture operates through a layered and integrated communication model. At the access layer, mobile users such as rescue teams, field workers and affected residents connect to the portable base station using wireless cellular communication. The portable base station provides temporary coverage and forwards the traffic to the emergency network core.

The portable COW contains the radio access equipment, antenna system, compact controller and independent power supply. The temporary network connects through satellite or microwave backhaul to the emergency command centre and any surviving external network. Mobile users connect directly to the temporary cell, while isolated locations form an ad hoc mesh and forward traffic through a gateway whenever a path is available.

4. Analysis of Cellular Network Components

Mobile Station (MS): The smartphone, radio terminal or connected device used by survivors and emergency personnel. It performs network discovery, registration, communication and mobility procedures.

Base Transceiver Station (BTS): Provides the wireless radio interface. In the portable deployment, compact antennas and radio units create temporary coverage over the affected area.

Base Station Controller (BSC): Manages radio resources, channel allocation, admission decisions and handoff control. In a compact deployment, these functions can be integrated into an emergency edge controller.

Mobile Switching Centre (MSC): Provides call control, switching and interconnection with external networks. It supports routing of emergency voice and signalling traffic.

Backhaul Gateway: Connects the emergency network to satellite, microwave or any surviving terrestrial network.

Power Unit: Uses battery and generator backup so that the communication system can operate independently of the damaged electrical grid.

5. Cellular Protocol Procedures

5.1 Registration / Network Attach

- The mobile device detects the broadcast signal from the portable BTS.
- The device selects the emergency cell and sends a registration or attach request.
- The network performs identity verification and authentication.
- Location and mobility information are updated in the temporary core.
- Radio resources and session parameters are allocated.
- The network confirms successful registration and the device becomes reachable.

5.2 Emergency Call and Data Session Setup

Emergency communication is handled using priority-based resource allocation. Rescue personnel, medical teams and command communication receive higher priority than ordinary traffic. After authentication, the network establishes the required bearer or session and forwards traffic through the available backhaul. Non-essential traffic can be limited when backhaul capacity is constrained.

5.3 Handoff Procedure

When a moving user experiences reduced signal quality, the mobile device measures neighbouring cells or available network segments and reports the measurements. The controller selects a suitable target, transfers the necessary context and switches the communication path. A make-before-break approach is preferred whenever possible to minimize interruption during critical communication.

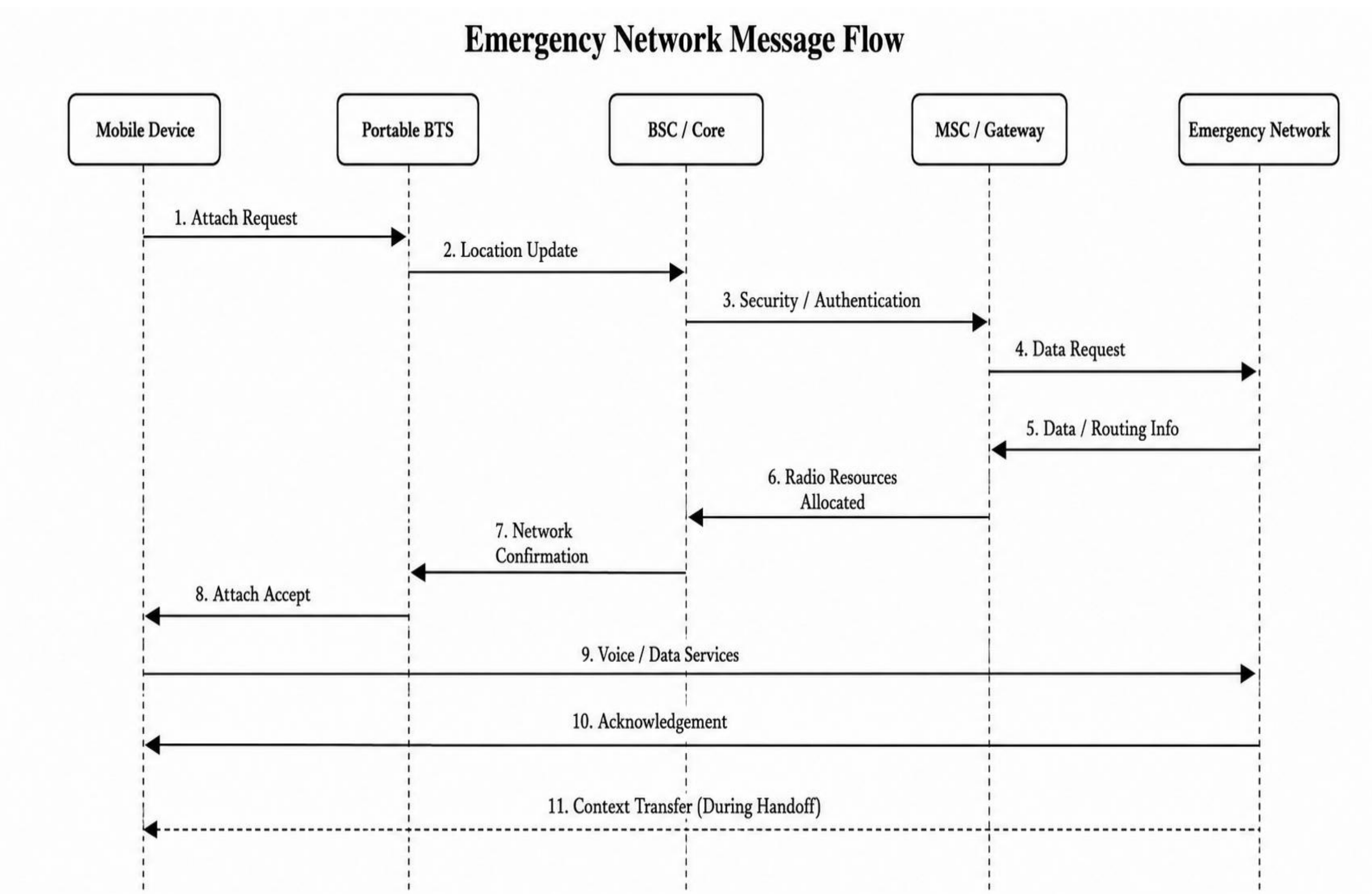


Figure 2. Registration, session setup and handoff message flow

6. Comparison of Multiple Access Techniques

Technique	Principle	Advantages	Limitations	Emergency Suitability
FDMA	Separate frequency band per user	Simple implementation	Low spectral efficiency	Useful for narrow legacy voice
TDMA	Users share a frequency using time slots	Predictable capacity	Requires synchronization	Suitable for basic legacy services
CDMA	Users share spectrum using unique codes	Interference tolerance and soft handoff	Complex power control	Useful in selected legacy deployments
OFDMA	Orthogonal subcarriers allocated dynamically	High efficiency and flexible scheduling	Higher processing complexity	Highly suitable for modern emergency broadband

Selection: OFDMA is preferred for a modern portable emergency network because radio resources can be dynamically scheduled according to priority. This allows critical voice, alerts and command data to receive resources before non-essential traffic.

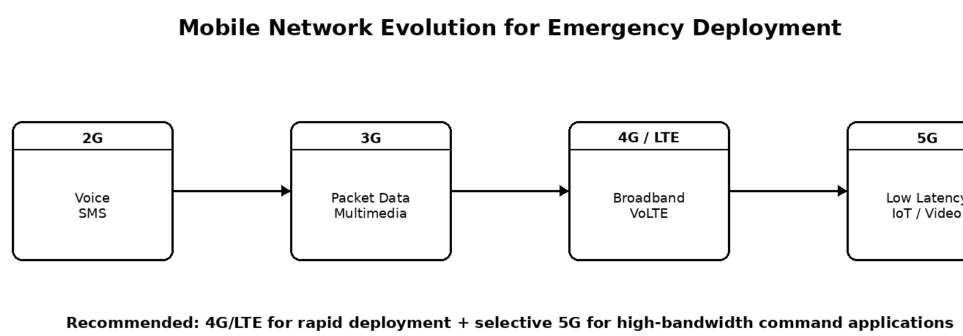
7. Mobile Network Evolution: 2G to 5G

ComparativeAnalysis:

The evolution from 2G to 5G shows a continuous improvement in data capacity, latency, network flexibility and service support. While 2G mainly supports voice and SMS communication, 3G introduced improved packet-based services. 4G/LTE provides high-speed broadband connectivity and is highly suitable for rapidly deployable emergency communication because of its wide device compatibility and mature infrastructure.

TechnologySelection:

For the proposed post-disaster scenario, a hybrid approach based mainly on **4G/LTE with optional 5G support** is recommended. LTE can provide stable voice and data connectivity quickly, while 5G can be deployed selectively near the emergency command centre.



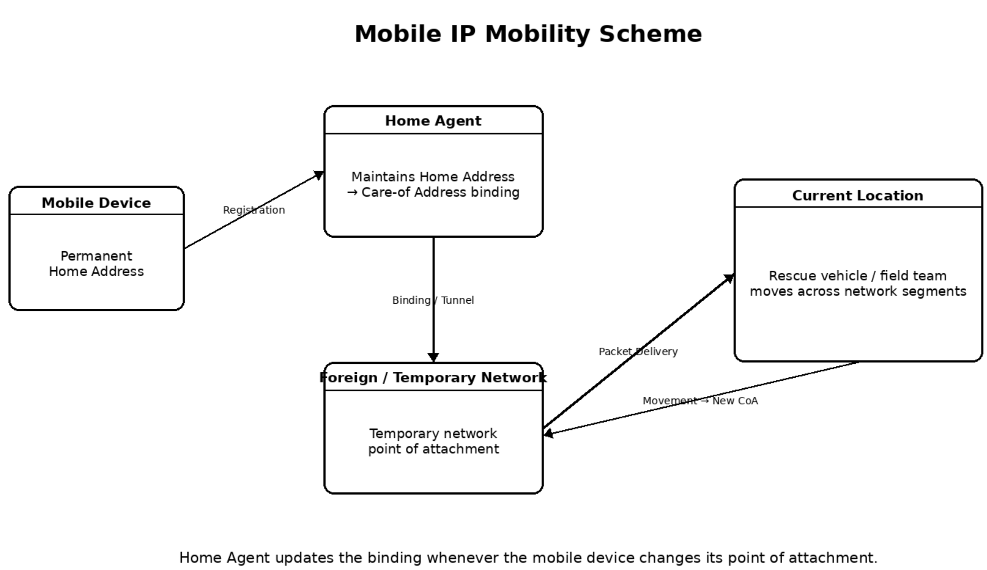
Generation	**Main Capability**	**Relevance to Emergency Deployment**
2G	Digital voice and SMS	Useful for low-rate alerts and compatibility
3G	Packet data and improved multimedia	Moderate relevance; many deployments are being phased out
4G/LTE	High-speed packet data and VoLTE	Strong practical choice for portable emergency deployment
5G	High capacity, low latency and advanced services	Useful for drones, video, command systems and critical IoT when resources are available

8. Recommended Emergency Technology Stack

A compact 4G/LTE Cell-on-Wheels is recommended as the primary deployment because LTE equipment is mature, widely supported and efficient for both voice and data. Optional 5G capability can be added near the command centre for high-bandwidth applications. SMS and emergency broadcast remain important because they operate with relatively low bandwidth requirements.

Requirement	Recommended Solution
Temporary coverage	Portable 4G/LTE Cell-on-Wheels with compact antennas
Backhaul	Satellite as primary option; microwave or surviving fibre as alternatives
Critical traffic	Priority QoS for responders and emergency communication
Public warning	SMS and emergency broadcast
Advanced services	Optional 5G small cell near the command centre
No-coverage zones	Wi-Fi / 802.11 ad hoc mesh
Power resilience	Battery + generator + optional solar support

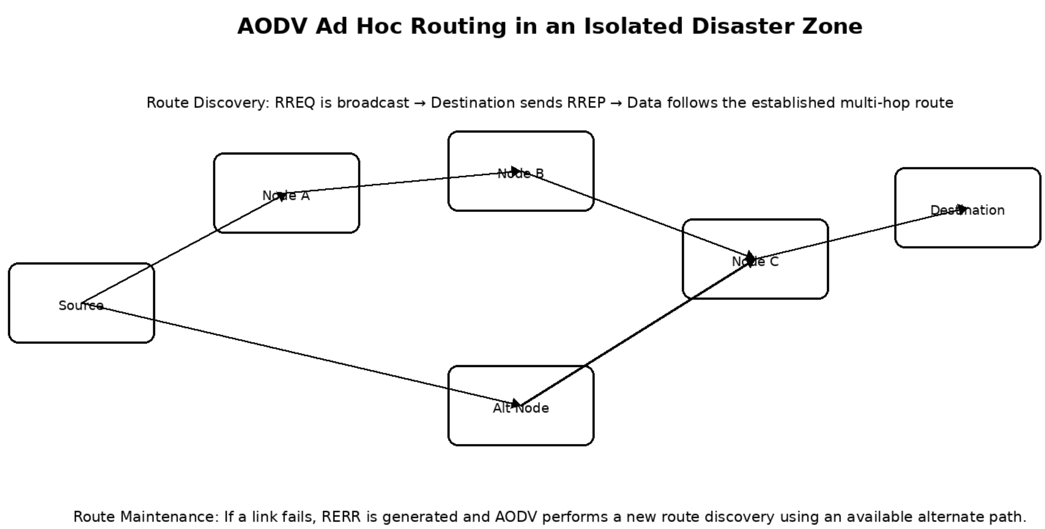
9. Mobile IP Based Mobility Scheme



Rescue vehicles and field teams may move between the temporary emergency cell and surviving network segments. Mobile IP maintains reachability by separating the device's permanent identity from its current point of attachment. The device retains a Home Address while a Home Agent maintains a binding to its current Care-of Address (CoA).

- A rescue device enters the disaster zone and detects the temporary or foreign network.
- The device obtains a Care-of Address (CoA) for the current network.
- A registration message updates the Home Agent with the new location.
- The Home Agent maintains the Home Address → CoA binding.
- Packets are forwarded or tunnelled to the current location of the mobile device.
- When the device moves again, a new registration updates the binding.

10. Infrastructure-less Ad Hoc Routing Scheme



Some villages and relief camps may remain outside the coverage of the portable cellular base station. In such areas, devices can form a self-organizing ad hoc network. AODV (Ad hoc On-Demand Distance Vector) is selected because it creates routes only when communication is required and can discover new paths after a link failure.

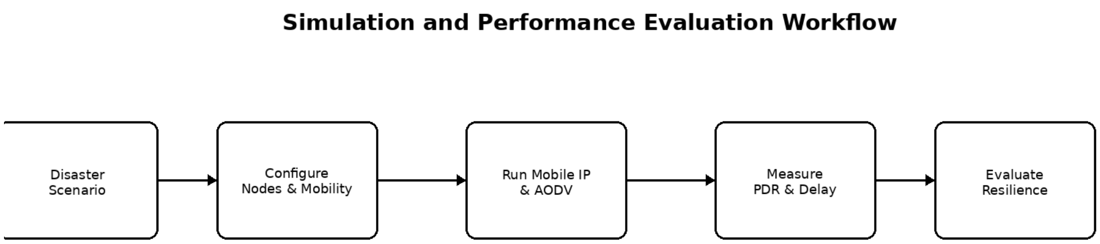
- Route Discovery: the source broadcasts a Route Request (RREQ).
- Route Formation: the destination or a suitable intermediate node returns a Route Reply (RREP).

• Data Forwarding: packets travel through the discovered multi-hop path.

• Route Maintenance: a Route Error (RERR) is generated when a link fails.

• Recovery: a new route discovery process attempts to establish an alternate path.

11. Simulation and Evaluation Methodology



The proposed design can be evaluated using NS-3, OMNeT++ or an equivalent wireless-network simulator. Two complementary scenarios are considered: Mobile IP movement between temporary and surviving network segments, and AODV multi-hop communication inside an isolated village or relief camp.

Parameter	**Illustrative Configuration**
Simulation area	5 km × 5 km disaster zone
Nodes	One portable-cell gateway and approximately 20–40 mobile/ad hoc nodes
Mobility	Rescue vehicles and responders moving between zones
Traffic	Emergency messages, periodic location updates and voice-like CBR traffic
Protocols	Mobile IP mobility management and AODV routing
Metrics	Packet delivery ratio, average delay, route recovery and partition tolerance
Failure model	Random node/link failures and intermittent backhaul availability

12. Performance Evaluation

The following table gives expected evaluation trends for the proposed design. These values serve as a design target and can be replaced by measured simulator results during implementation.

Metric	**Mobile IP Scheme**	**AODV Ad Hoc Scheme**	**Interpretation**
Packet Delivery Ratio	≈ 96%	≈ 90–94%	Managed infrastructure generally provides more stable delivery
Average End-to-End Delay	≈ 70–120 ms	≈ 100–220 ms	Multi-hop forwarding and route discovery increase delay
Mobility Support	High	Moderate to High	Mobile IP directly manages network attachment changes
Partition Resilience	Moderate	High for local clusters	Mesh can continue local communication without backhaul
Recovery	Fast after binding update	Depends on route discovery	AODV reconstructs routes after link failure

13. Reliability, Delay and Network Partition Resilience

Reliability: The hybrid design provides redundancy through portable cellular access, alternative backhaul and local ad hoc forwarding. A failure in one component does not necessarily terminate all communication.

Delay: Priority QoS is used for voice and command traffic. The cellular path provides low-delay communication where coverage exists, while the ad hoc network accepts additional delay in exchange for connectivity in isolated areas.

Network Partition Resilience: If external backhaul is lost, local communication inside a relief camp or village can continue through the ad hoc mesh. Gateway nodes can synchronize queued information with the command centre after connectivity is restored.

14. Design Decisions and SDG Relevance

Hybrid Architecture: A single communication technology cannot guarantee connectivity in all disaster conditions. The cellular layer provides managed coverage and QoS, while ad hoc networking provides communication where infrastructure is absent.

SDG 9 – Industry, Innovation and Infrastructure: The design supports resilient, rapidly deployable and adaptable communication infrastructure.

SDG 11 – Sustainable Cities and Communities: Reliable emergency communication improves community resilience and enables faster rescue, evacuation and relief coordination.

SDG 13 – Climate Action: Climate-related disasters require stronger preparedness and response systems. Independent power support and resilient communication improve the ability to respond to extreme events.

15. Reflection and Learning Outcomes

A major design challenge was balancing rapid deployment with performance. Advanced technologies such as 5G offer high capacity but may require more complex equipment and stronger backhaul. Therefore, LTE was selected as the practical foundation, with optional 5G support for advanced applications.

Another challenge was maintaining communication in completely isolated locations. This requirement led to the integration of an AODV-based ad hoc mesh. The work demonstrates that Mobile IP and ad hoc routing solve complementary mobility problems: Mobile IP maintains reachability across managed networks, while ad hoc routing creates communication paths when infrastructure is unavailable.

16. Conclusion

The proposed rapidly deployable post-disaster communication network combines a portable Cell-on-Wheels, resilient backhaul, priority-based traffic management, Mobile IP mobility support and an AODV-based ad hoc mesh. OFDMA-based 4G/LTE provides a practical foundation for temporary emergency coverage, while optional 5G can support advanced command applications. The hybrid approach improves reliability, controls communication delay and provides resilience when parts of the network become partitioned.

17. GitHub Deliverables

- Technical report PDF
- Network architecture and message-flow diagrams
- Mobile IP mobility scheme design or simulation files
- AODV ad hoc routing design or simulation files
- Simulation configuration and evaluation results
- README describing assumptions, setup and performance metrics

Suggested Repository Structure

```
Mobile-Computing-Disaster-Network/
├── report/
└── ITA03_Mobile_Computing_Assignment.pdf
├── diagrams/
│   ├── architecture.png
│   └── message_flow.png
├── simulation/
│   ├── mobile_ip/
│   └── ad_hoc_aodv/
├── results/
│   └── evaluation_results/
└── README.md
```

18. References

- Perkins, C. E., & Belding-Royer, E. M. (1999). Ad hoc On-Demand Distance Vector Routing. *Proceedings of the 2nd IEEE Workshop on Mobile Computing Systems and Applications*, 90–100. DOI: 10.1109/MCSA.1999.749281.
- Perkins, C. E., Belding-Royer, E. M., & Das, S. R. (2003). Ad hoc On-Demand Distance Vector (AODV) Routing. *RFC 3561*.
- Royer, E. M., & Perkins, C. E. (1999). Multicast Operation of the Ad-hoc On-Demand Distance Vector Routing Protocol. *Proceedings of MobiCom '99*, 207–218. DOI: 10.1145/313451.313538.
- Das, S. R., Perkins, C. E., & Royer, E. M. (2000). Performance Comparison of Two On-demand Routing Protocols for Ad Hoc Networks. *Proceedings of IEEE INFOCOM*, 3–12.
- Sanmateu, A., Paint, F., Morand, L., Tessier, S., Fouquart, P., Sollund, A., & Bustos, E. (2002). Seamless mobility across IP networks using Mobile IP. *Computer Networks, 40*(1), 181–190. DOI: 10.1016/S1389-1286(02)00273-6.
- Cox, C. (2012). Orthogonal Frequency Division Multiple Access. In *An Introduction to LTE*. Wiley. This is particularly useful for your OFDMA/LTE discussion.
- Layered OFDMA and Its Radio Access Techniques for LTE-Advanced. (2009). *IEICE Transactions on Communications, E92.B*(5), 1743–1751. DOI: 10.1587/transcom.E92.B.1743.
- Royer, E. M., & Toh, C.-K. (1999). A Review of Current Routing Protocols for Ad-Hoc Mobile Wireless Networks. *IEEE Personal Communications, 6*(2), 46–**55.**